

Probabilistic model checking

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Motivation

We want to model a probabilistic system and check its properties.

Examples include:

- Randomised distributed algorithms
- Communication, network and multimedia protocols
- Security protocols
- Biological systems
- Planning and synthesis
- Performance and reliability
- Game theory
- Power management
- ...

How do we model a probabilistic system?

1. describe states and transitions
2. make number of state finite by abstraction
3. we assign probabilities to transitions

Modeling customer on a shopping website

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Discrete-time Markov chains

The Myriad of Models

- Discrete-time Markov chains
- Continuous-time Markov chains
- Markov decision processes
- Markov automata
- Probabilistic timed automata
- ...

Differences between models?

Features:

- Discrete vs continuous time
- Nondeterminism
- Rewards
- intervals
- ...

Continuous vs Discrete

Continuous-time Markov chains

Continuous-time Markov chains

Continuous-time Markov chains

